I.E N5 CHAPTER 01 TUT 2022

1.) A serial RLC circuit which is connected to a 250V AC supply, consists of 0,15H, a resistor of 10Ω and a capacitor of 100µF. Calculate the total impedance, total current and the Q-factor, given that the supply frequency is 50Hz.

2.) A serial circuit consists of a 12Ω resistor, a 150µH inductor and an adjustable capacitor. This circuit is connected to a 10V,50Hz sine wave, Calculate the following:

a.) The value of the capacitor to let the circuit resonate at 50kHz
b.) The total current through the circuit
c.) The voltage drop across each component
d.) The reactance X_L and X_C
e.) The total impedance and the Q-factor
f.) Draw the phase diagram of the voltage drops in this resonant circuit.
g.) Calculate the total impedance
h.) Calculate the voltage drops across the resistor, inductor and capacitor of a serial circuit with the following values: R=5Ω, X_L=j40Ω, X_C =-j80Ω, $V_T = 30V$
i.) Draw the phasor diagrams of the impedances and voltage drops.

3.) An inductor of 5mH is connected in series with an adjustable capacitor. At which value must the capacitor be tuned in order to tune the circuit at 300kHz?

4.) A serial circuit consists of a resistor of 20Ω, an inductor of 10mH, a capacitor of 10µF and a supply voltage of 250V,50Hz.Calculate the voltage drop across each component and draw the phase diagram, Calculate the value at which the capacitor must be tuned in order to let the circuit resonate.

5.) A capacitor is connected to a 110V ,60Hz supply. if the current flow is 250mA, what is the capacitance of the capacitor?

6.) The following 50Hz impedances are given:

$Z_1 = 3 + j10\Omega$, $Z_2 = 5 - j16\Omega$

a.) What is the total impedance of the form if $Z_1$and Z_2 is connected in series?
b.) What is the total impedance of the form if $Z_1$and Z_2 is connected in parallel?

7.) A parallel RL circuit with j$X_L = j15\Omega$ and R=20Ω:

a.) Calculate the current through R and L as well as the total current with a voltage supply of 20V, 800Hz?
b.) Draw a phase diagram of the currents.

8.) Draw the phase diagram of the currents if the inductor in question 16 is replaced by a capacitor with a reactance of –j15Ω.

9.) A Coil with a 20milli Henry inductance and a DC resistance of 5Ω is connected to an adjustable capacitor in parallel. The combinations is connected to a 5kΩ resistor in series :

a.) Calculate the resonant frequency with C tuned to 10μF.
b.) Calculate the dynamic impedance.
c.) Calculate the Q-factor of the circuit

10.) A serial RL circuit with values of j31,4Ω and 2Ω is connected across a capacitor in parallel. If the supply voltage is 100V, 500Hz:

a.) Calculate the value of the capacitor at resonance
b.) Calculate the current through each branch.
c.) Calculate Z_d
d.) Calculate the Q-factor.[1]

11. It takes 1 ms to reach 63,2% of the total current (IT) in a serial RL circuit if $R = 10\ \Omega$ and $I_T = 0{,}5$ A. Calculate each of the following:

11.1 The value of V_L

11.2 I_L after 1ms